

Protection against radiation oxidative damage in mice by Triphala

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Abstract

Protection against whole body γ -irradiation (WBI) of Swiss mice orally fed with Triphala (TPL), an Ayurvedic formulation, in terms of mortality of irradiated animals as well as DNA damage at cellular level has been investigated. It was found that radiation induced mortality was reduced by 60% in mice fed with TPL (1 g/kg body weight/day) orally for 7 days prior to WBI at 7.5 Gy followed by post-irradiation feeding for 7 days. An increase in xanthine oxidoreductase activity and decrease in superoxide dismutase activity was observed in the intestine of mice exposed to WBI, which, however, reverted back to those levels of sham-irradiated controls, when animals were fed with TPL for 7 days prior to irradiation. These data have suggested the prevention of oxidative damage caused by whole body radiation exposure after feeding of animals with TPL. To further understand the mechanisms involved, the magnitude of DNA damage was studied by single cell gel electrophoresis (SCGE) in blood leukocytes and splenocytes obtained from either control animals or those fed with TPL for 7 days followed by irradiation. Compared to irradiated animals without administering TPL, the mean tail length was reduced about three-fold in blood leukocytes of animals fed with TPL prior to irradiation. Although, similar protection was observed in splenocytes of TPL fed animals, the magnitude of prevention of DNA damage was significantly higher than that observed in leukocytes. It has been concluded that TPL protected whole body irradiated mice and TPL induced protection was mediated through inhibition of oxidative damage in cells and organs. TPL seems to have potential to develop into a novel herbal radio-protector for practical applications.

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1. Introduction

Triphala (TPL) is one of the most popular Ayurvedic medicines, primarily used as a laxative and an intestinal tonic. In addition, TPL has many other medical uses,

which requires understanding of the mechanisms of its therapeutic effects. This formulation, which is a mixture of dried and powdered fruits of three plants, *Terminalia chebula*, *Phyllanthus emblica* and *Terminalia bellerica* in equal proportions, has been reported to possess nutritional properties, blood cleansing and liver protective properties [1–4]. Interestingly, TPL is also shown to possess anticancer activities [5,6]. Recently, it was reported that intra-peritoneal administration of TPL significantly modified γ -radiation induced mortality in mice

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[7]. However, mechanisms of the radiation protective action of TPL is lacking in literature.

In the present study, TPL was fed to mice orally to examine its radio-protective property as it may offer a more suitable route of administration for practical applications [8]. Moreover, we have examined the possible mechanisms of action of this drug at the intestinal level, which is a sensitive target for radiation damage after WBI and have evaluated the magnitude of DNA damage in leukocytes and spleen cells. Studies on laboratory animals have shown that the mechanisms of TPL induced protection of mice against both gastrointestinal and hemopoietic death was involving free radical mechanism [9]. However, not much is known about TPL modulation of the activities of radiation-sensitive enzymes that play a role in the induced mortality. The present study describes the modulation of enzymes, such as, XO and SOD in the intestine of mice exposed to WBI with or without oral feeding of TPL. Results have shown that orally fed TPL was capable of significantly preventing γ -radiation induced mortality of mice exposed to WBI, which was partly attributable to inhibition of cellular oxidative radiation damage processes.

2. Materials and methods

2.1. Materials

Triphala is a mixture of dried and powdered fruits of three plants, *T. chebula*, *P. emblica* and *T. bellerica* in equal proportions (1:1:1). Triphala extract in the form of powder was obtained from Ajanta pharmaceuticals, Mumbai. Required quantity of extract powder was freshly dissolved in double distilled water and filtered through 0.22 μ m syringe filter (Milipore, USA) and used for the experiments. Dulbecco's Modified Eagle's Medium (DMEM) was purchased from Gibco Co., USA and fetal bovine serum (FBS) was obtained from Biomed, France. Agarose, Na₂-EDTA, Triton X-100, DMSO, Tris-HCl and SYBR Green II were obtained from Sigma Chemicals Inc., St. Louis, MO, USA, NaCl from S.D. Fine Chemicals (India), NaOH from Thomas Baker Chemicals Ltd. (India) and fully frosted slides from Erie Scientific Co., USA.

2.2. Animals

All animal experiments were conducted adhering to the guidelines from Institutional Animal Ethics Committee (IAEC). Six to eight weeks old female Swiss mice weighing 20–25 g, bred at the animal house facility of the centre and maintained under standard conditions of temperature and humidity were used in the study. Animals were provided with food and water *ad libitum*. Five animals were housed in a polypropylene cage containing sterile paddy husk as bedding.

2.3. TPL administration and irradiation of mice

Inbred, female Swiss mice were divided into 4 groups with 10 animals in each group and were treated as follows:

1. TPL: The animals of this group received 1 g/kg body weight/day Triphala, orally for 14 days.
2. DDW + irradiation: The animals of this group were given equal volume of sterile double distilled water instead of TPL as described for group 1 and were subjected to whole body γ -irradiation of a single lethal dose of 7.5 Gy on the 7th day.
3. TPL + irradiation + TPL: Mice fed with 1 g/kg body weight/day Triphala for 14 days, starting from 1 week before radiation and continued a week later after a single dose of whole body γ -irradiation of 7.5 Gy.
4. Irradiation + TPL: Mice fed with 1 g/kg body weight/day TPL for 7 days, after whole body γ -irradiation of 7.5 Gy.
5. Irradiation was done using a Theratron 780E machine (Nuclear Enterprises, UK, 1.522 Gy/min (8.6 min) having a γ -ray emitting cobalt 60 source.

2.4. Effect of TPL and radiation on intestinal xanthine oxidoreductase (XO) and superoxide dismutase (SOD) activity

The assay procedure followed for the determination of XO system was essentially the same that of described by Kaminski and Jezewska [10]. Briefly, the enzyme activity was monitored spectrophotometrically by measuring uric acid formation at 293 nm with saturated concentration of xanthine (20 μ M) as the substrate in 1 ml of 50 mM phosphate buffer, pH 7.5. In all cases the reaction was started by the addition of 12 nM of XO. One unit of xanthine oxidoreductase is defined as the amount of the enzyme that produces 1 nmol of uric acid in 1 ml of assay volume per min at 25 °C. Xanthine dehydrogenase activity was determined by measuring the appearance of NADH at 340 nm in a Hitachi Double Beam Spectrophotometer with NAD present as cofactor. The assay mixture contained 50 mM phosphate buffer, pH 7.4, 100 μ M xanthine and an aliquot of enzyme sample with and without NAD (100 μ M) in a final volume of 3 ml. XO activity of the tissue preparations were followed at 290 nm by measuring the amount of uric acid formed in the presence of O₂ without the added NAD. The enzyme activity of dehydrogenase/oxidase is expressed in units/mg protein. One unit of enzyme activity is defined as the amount of enzyme required to oxidise 1 μ M of xanthine per min. Protein content of the tissue preparations were determined by following the method described by Lowry et al. [11].

SOD activity was determined spectrophotometrically by measuring the autooxidation of epinephrine at 480 nm [12]. The assay mixture contained 50 mM sodium carbonate buffer (pH 10.2) containing 0.10 M EDTA and an aliquot of sample. The reaction was initiated with 30 mM epinephrine. The rate of assay was measured at 480 nm for 180 s in Hitachi Double Beam spectrophotometer. SOD activity was expressed as

units (U) of SOD/mg protein where 1 unit of SOD is defined as the amount of enzyme present that inhibits auto oxidation of epinephrine by 50%.

2.5. Estimation of DNA damage by alkaline single cell gel electrophoresis

Alkaline single cell gel electrophoresis (SCGE) was conducted as described earlier [13–15]. Cell viability was tested by Trypan Blue dye exclusion test in 0.02% Trypan Blue. Spleen cells were washed with phosphate buffered saline and resuspended in the same to get a cell density of $\sim 1 \times 10^5$ cells in 10 μ l. Briefly, 100 ml of heparinized whole blood samples were mixed with 1.5 ml of 0.8% agarose solution at 42 °C and the mixture was poured on fully frosted slides uniformly. After solidification, the slides were immersed in lysing solution, (2.5 M NaCl, 100 mM Na₂-EDTA with freshly added 1% Triton X-100 and 10% DMSO), for 1 h at 4 °C. The slides were removed from the lysing solution, washed with alkaline electrophoresis buffer three times and then placed on horizontal electrophoresis tank filled with freshly prepared alkaline buffer (300 mM NaOH, 1 mM Na₂-EDTA, pH 13.0) at 4 °C. The slides were kept in the same buffer (pH 13) for 20 min to allow DNA unwinding and conversion of alkali-labile sites to single strand breaks. Electrophoresis was carried out for 20 min at 1.1 V/cm (25 V, 300 mA) using a compact power supply. After electrophoresis, the slides were washed gently to remove the alkali and detergents by placing them horizontally and flooding them slowly with 0.4 M Tris buffer at pH 7.5. The slides were stained in 50 μ l of diluted SYBR Green II diluted 1:10,000 in 10 mM Tris-HCl, pH 7.5 and 1 mM EDTA. The slides were viewed by fluorescence microscopy with a 515–560 nm excitation filter and analyzed with the help of a Digital Imaging System. (SCGE-Pro) developed in house at Bhabha Atomic Research Centre, Mumbai, India [14]. The experiment was done in triplicate and a minimum of 50 cells were scored per slide.

Depending on the type and intensity of the stress, DNA strands break into fragments that migrate towards the anode during electrophoresis. The extent of DNA damage is then quantified using parameters such as tail length, tail moment and percent of DNA in tail. Tail length of the comet indicates the amount of damage as longer tails indicate that the strand breaks are more frequent and that the DNA is fragmented to several smaller molecules. Percent DNA in the tail gives the percentage of the DNA in the tail of the comet taking the total DNA of the cell as 100%. Tail moment is a commonly accepted unit of DNA damage, which normalizes the difference in the size of the nucleus studied, e.g. blood leukocytes [16]. It is the product of fraction of the DNA in the tail of the comet and the tail length.

2.6. Statistical analysis

Wherever necessary, the one-way ANOVA test was performed using Origin (version: 6.1) software. All *in vitro* exper-

iments were done in triplicates and each data point represents the average of at least three independent experiments.

3. Results

3.1. Effect of TPL on γ -radiation induced mortality in mice

Studies were carried out to examine the modification of mortality after whole body γ -radiation exposure (7.5 Gy; LD50₃₀) of mice by TPL. Mice were given WBI with or without feeding TPL as described in Section 2. Most animals in the DDW + irradiation (irradiated control) group showed signs of radiation sickness such as weakness, epilation, ruffling of hair, etc. within a week after radiation exposure. The first mortality in this group was observed on the 5th day and 60% animals died within 30 days after irradiation (Fig. 1) resulting in only 40% survival. Compared to this group, none of the animals in TPL + irradiation + TPL protocol group, showed any signs of radiation sickness during post-irradiation 30 day observation period. These animals showed 100% survival during extended post-irradiation period of 50 days. Thus, TPL (1 g/kg body weight/day) when fed before and

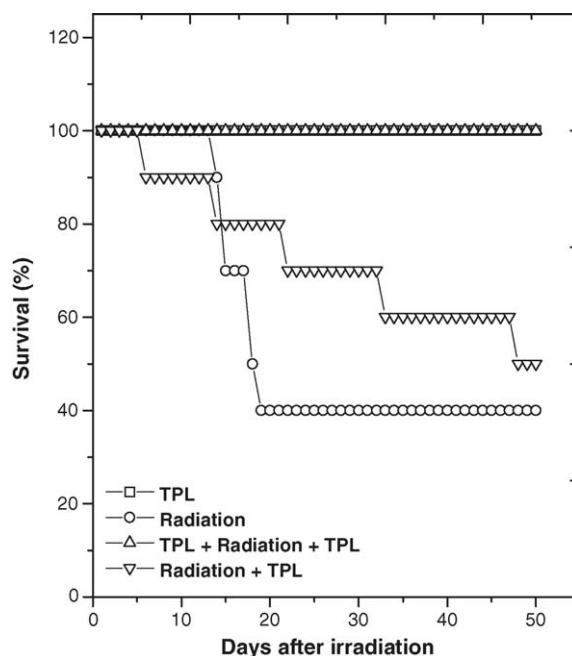


Fig. 1. Effect of TPL (1 g/kg body weight/day) on radiation (7.5 Gy) induced mortality in mice exposed to γ -radiation. Six to eight weeks old female Swiss mice were divided into 4 groups with 10 animals in each group and treated in the following manner: mice fed with TPL for 14 days (□); mice fed with distilled water and irradiated (○); mice fed with TPL for 14 days, 7 days before and 7 days after irradiation (△) and mice fed TPL for 7 days after irradiation (▽).

after irradiation, significantly reduced radiation-induced mortality by 60% in these mice. However, when irradiated mice were fed with TPL (irradiation + TPL group), for 7 days, the survival by day 30 post-irradiation was only 70% showing a relative reduced protection by the drug. It was further observed that all animals in unirradiated control group fed with the same amount of TPL for 14 days did not show any signs of toxicity during the observations.

3.2. Modulation of enzyme activities in the mice intestine fed with TPL followed by WBI

The changes in activities of enzymes of whole body irradiated mice were examined with or without TPL feeding. It was found that compared to the un-irradiated control, whole body radiation exposure of mice caused an increase in XO activity by more than two-fold (Fig. 2a). However, mice fed with TPL for 7 days prior to irradiation, showed significant reduction in XO activity.

It was further observed that the activity of intestinal SOD decreased as compared to that of the control (Fig. 2b). However, mice fed with TPL did not show any significant reduction in SOD enzyme activity after irradiation.

3.3. TPL prevents radiation induced cellular DNA damage

Studies were carried out to evaluate the oxidative damage to cellular DNA in mice with or without TPL feeding, following irradiation. For this purpose, mouse blood leukocytes and spleen cells from different groups of animals were subjected to single cell gel electrophoresis and DNA damage was analyzed by using SCGE-pro software. Typically, nucleoids of untreated control cells stained with SYBR-Green were observed to be circular in shape while those from irradiated cells looked like comets with fluorescence intensity maximum in the head region and diminishing towards the tail region, reflective of higher magnitude of DNA damage. The frequency distribution histograms of tail length (TL), tail moment (TM) and percent DNA (TDNA) in tail of blood leukocytes and spleen cells taken from mice non-treated (A–C), TPL alone (D–F), 7.5 Gy (G–I) and fed with TPL before and after irradiation (J–L) are shown in Fig. 3a and b respectively. As evident from figures, compared to cells from sham-irradiated mice, the values for tail length, tail moment and percent DNA in tail increased for both these cells in mice exposed to a dose of 7.5 Gy (Fig. 3a and b; G–I). The response to TPL and radiation treatment in blood leukocytes followed a pattern

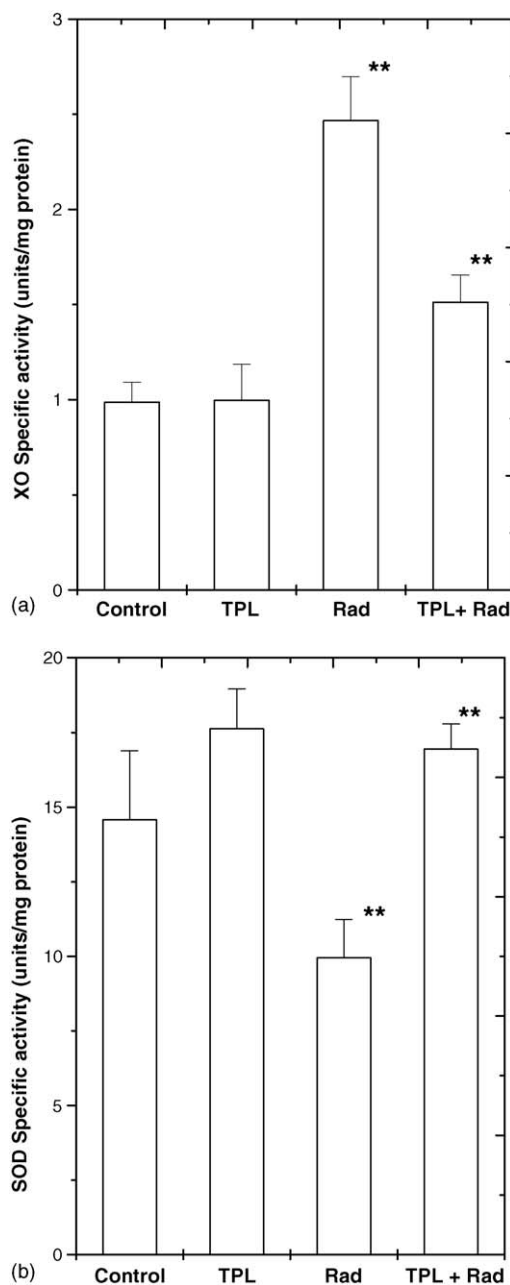


Fig. 2. Effect of TPL and radiation on enzyme activity profile of xanthine oxidoreductase and superoxide dismutase. Swiss mice were divided into four groups with five animals in each group and treated in the following manner: untreated mice (control); mice fed with Triphala for 7 days (TPL); mice fed with distilled water for 7 days and exposed to whole body γ -irradiation of 7.5 Gy (Rad.) and mice fed TPL for 7 days and exposed to whole body γ -irradiation of 7.5 Gy on the 7th day. Animals were killed by cervical dislocation after 7 days of treatment and intestines removed 1 h. post-irradiation. Tissues were then processed for xanthine oxidase assay (a) and superoxide dismutase assay (b) as described in Section 2. Values for radiation group is significantly higher from that of TPL + radiation ** $p < 0.01$.

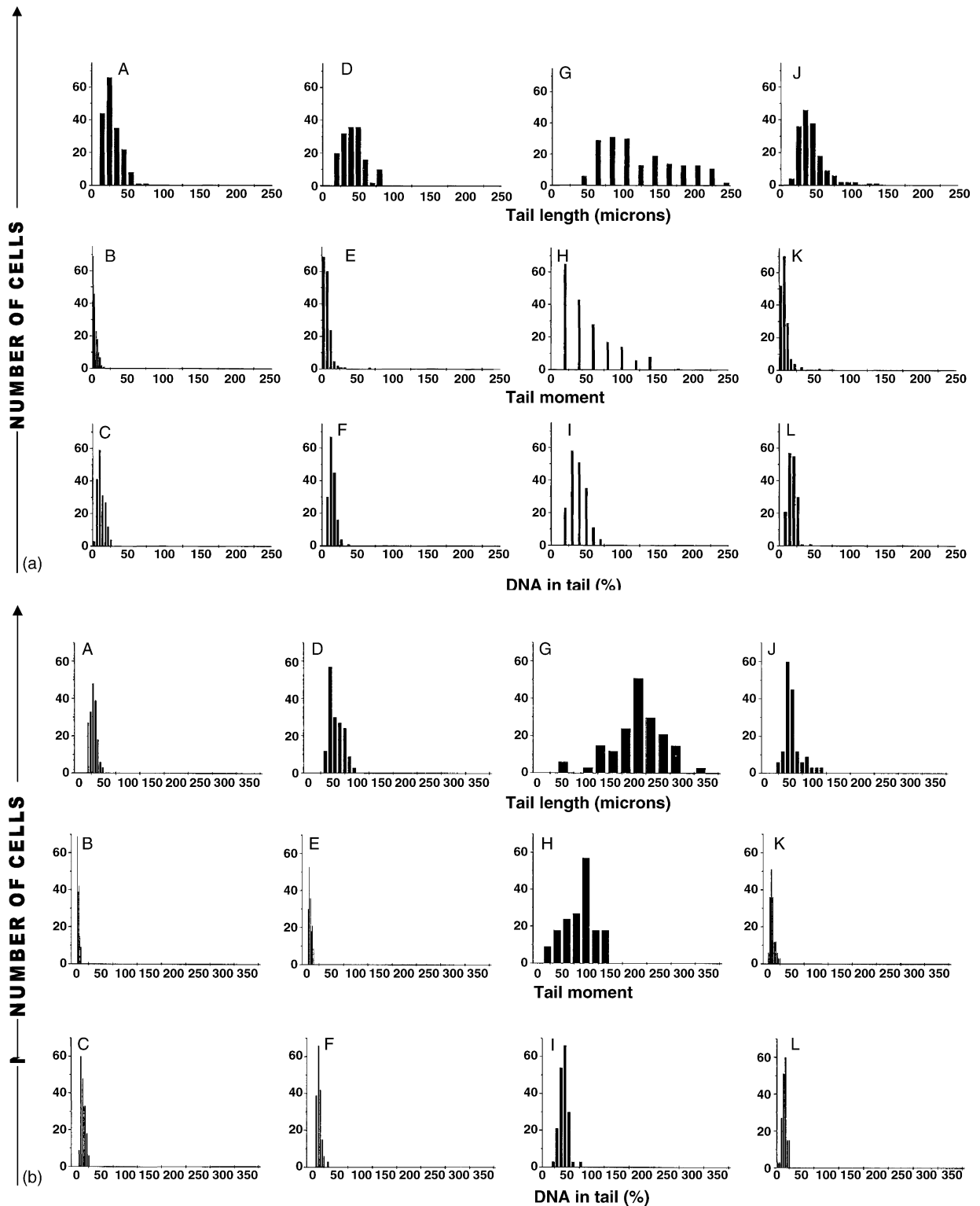


Fig. 3. Frequency distribution pattern of tail length, tail moment and percent DNA in tail of mouse blood leukocytes (a) and splenocytes (b): (A) tail length; (B) tail moment; (C) percent DNA in tail of un-irradiated mice; (D) tail length; (E) tail moment; (F) percent DNA in tail of mice fed Triphala (1 g/kg body weight/day for 14 days); (G) tail length; (H) tail moment; (I) percent DNA in tail of whole body irradiated mice at 7.5 Gy; (J) tail length; (K) tail moment and (L) percent DNA in tail of whole body irradiated mice at 7.5 Gy, after feeding Triphala (1 g/kg body weight/day for 14 days).

similar to spleen cells. However, the relative magnitude of DNA damage after radiation exposure was found to be higher in splenocytes (Fig. 3b; G–I) than leukocytes (Fig. 3a; G–I). Cells from the group of mice fed with TPL followed by irradiation showed significant inhibition in DNA damage measured in terms of TL, TM and TDNA (Fig. 3a and b; J–L). In addition, the pattern and width of distribution of DNA damage assay parameters for cells from mice fed only with TPL (Fig. 3a and b; D–E) followed nearly similar trend as control (Fig. 3a and b; A–C). It was interesting to find that width of distribution was significantly wide in these parameters, and the median value of distribution shifted to right in cells

from irradiated mice (Fig. 3a and b; G–I). When, animals were fed with TPL, the width of distribution and the magnitude of DNA damage were found restored nearly similar to that of sham-irradiated control (Fig. 3a and b; A–C). For comparisons, the magnitude of DNA damage was further scored for the four experimental groups and these data are plotted in Fig. 4a and b for parameters TL (i), TM (ii) and TDNA (iii). For the radiation dose, mean TL for leukocytes was 120 μm , while for splenocytes it was $\sim 200 \mu\text{m}$. Similarly, the tail moment values in samples from irradiated mice were about 60% higher in splenocytes than leukocytes. TL for control was $29 \pm 7 \mu\text{m}$ while for radiation treated it was $206 \pm 55 \mu\text{m}$

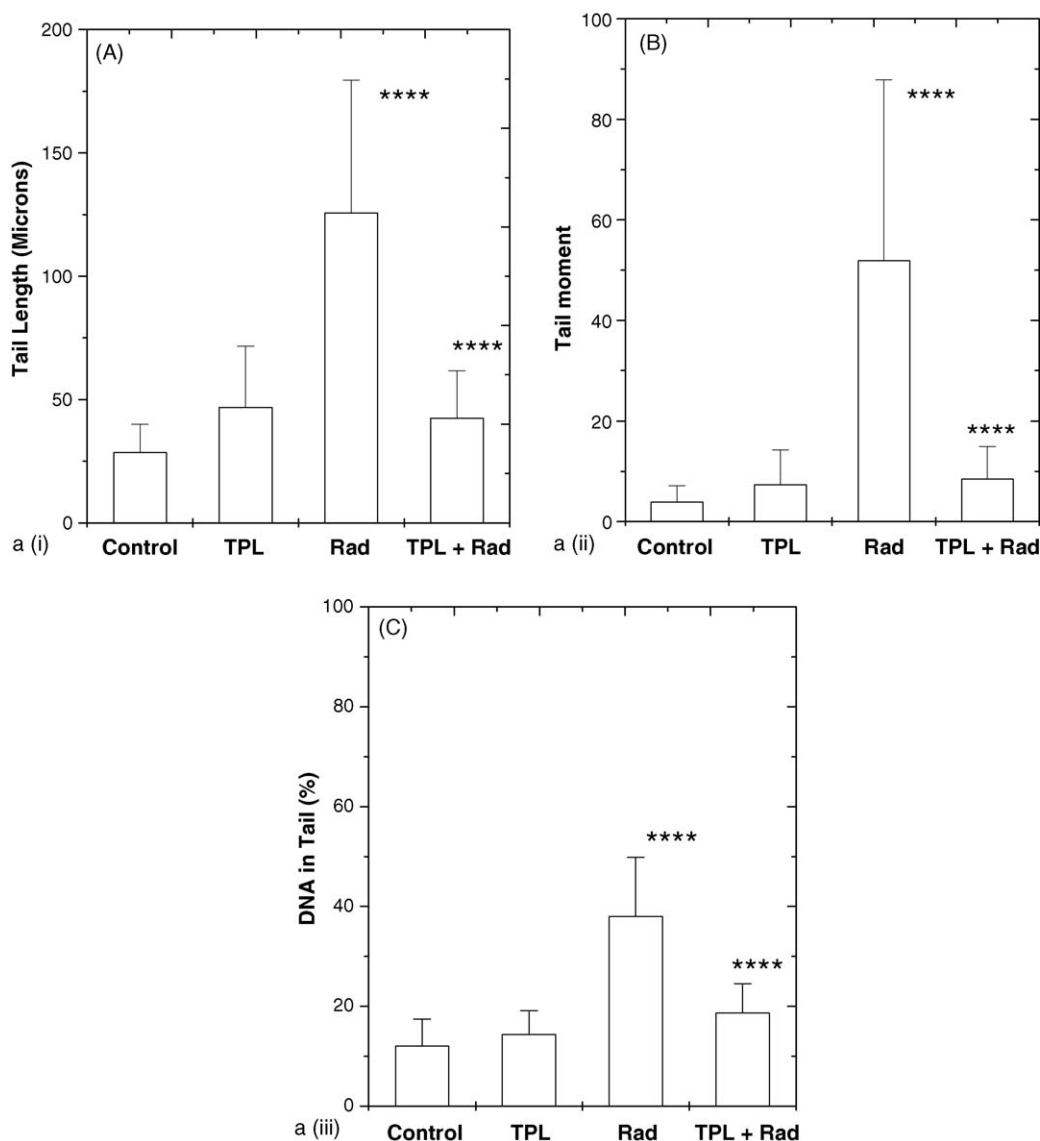


Fig. 4. Single cell gel electrophoresis analysis of DNA damage in mouse blood leukocytes (a) and splenocytes (b) of TPL and radiation treated mice: comparison between the different experimental groups using parameters: (i) tail length; (ii) tail moment and (iii) percent of DNA in tail.

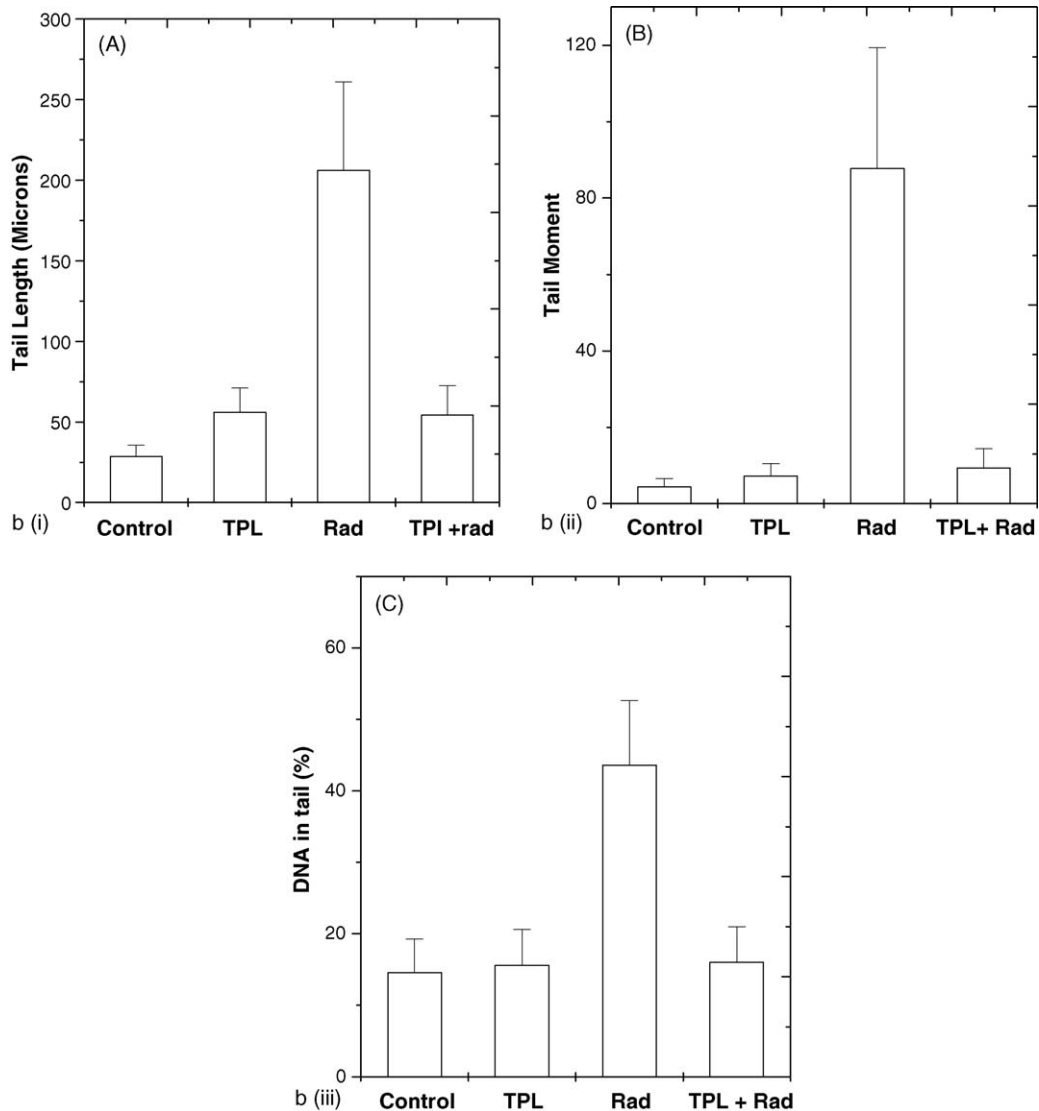


Fig. 4. (Continued).

in splenocytes. However, for the irradiated group treated with TPL, TL was 54 ± 18 (Fig. 4a; i). The results also indicate that the inhibition in radiation-induced DNA damage by TPL was relatively more efficient in spleen cells (74%) compared to leukocytes (66%). Although, it was seen that TPL treatment alone induced some DNA damage, it was not significantly different from that of control. In case of TM measurements, mean $TM \pm SD$ for mouse blood leukocytes was 4 ± 3.2 in the control group, while for the irradiated group it was 52 ± 36 , which was restored to 8.4 ± 6.5 in irradiated, TPL treated group (Fig. 4a; ii). The large standard deviation in the radiation group might be associated with differences in sensitivity to radiation in the different types of blood

cells. Spleen cells were found to show a similar trend (Fig. 4b; ii). TM results also showed that the protective effect of TPL was marginally higher in spleen cells (89%) than in blood leukocytes (84%). Analysis of percent DNA in tail of mouse blood leukocytes and spleen also showed similar trend as in TL and TM (Fig. 4a; iii and b; iii). These results showed that TPL remarkably prevented radiation-induced DNA damage and protected cells.

4. Discussion

In the previous reports from our laboratory, we have shown the novel anti-tumor property of TPL in human

cancer cell lines and possible underlying mechanisms were evaluated [5,6]. *In vitro* and *in vivo* studies had shown that TPL killed cancer cells by an ROS dependent apoptotic mechanism. It was further shown that TPL neither increased ROS nor toxicity in normal cells [5]. Since, the components of TPL are known to possess anti-oxidant properties [17–21], we were interested in exploring the radio-protective abilities of TPL and the underlying mechanisms of action at cellular level.

Our results showed that oral feeding of TPL (1 g/kg body weight/day) produced a profound effect in protecting mice against radiation-induced mortality (Fig. 1). It is interesting to note that the protection of mice by TPL against radiation could significantly be achieved in an oral feeding regime. These results have corroborated earlier reports [7]. However, unlike the earlier study, where TPL was administered to the experimental animals by intra-peritoneal injection, we chose a more convenient route, i.e. oral administration, as TPL is known to have least side effects and toxicity.

Since intestine is one of the major targets for radiation induced oxidative damage, the alterations in the levels of free radical generating and antioxidant enzymes are associated with the appearance of gastro-intestinal syndrome and eventually mortality. The biological effect of ionizing radiation is the net effect of free radical reactions and action of a range of cellular defense machinery. To further understand the radio-protective ability of TPL, we have evaluated the activities of XO and SOD, two important enzymes which are known to be affected by radiation exposure [22–24]. It is known that XO catalyzes the oxidation of xanthine to uric acid and generates superoxide ($O_2^{\bullet-}$) and hydrogen peroxide (H_2O_2), respectively. Superoxide ($O_2^{\bullet-}$) the byproduct of this reaction needs to be neutralized by the cellular defense enzyme, i.e. SOD. The level of SOD in our experimental condition has been found altered. Earlier studies have shown that mice treated with SOD showed substantial decrease in oxidative stress and attenuation of radiation-induced intestinal inflammatory response [25]. Oral feeding of TPL, as was done in the current study, seems to have affected the intestinal cells resulting in the modulation of antioxidant enzymes like SOD. We have found increase in XO activity, but decrease in SOD activity in mice intestine after radiation exposure, which lends support to earlier studies [22–24]. However, the effect was reverted towards control when animals were fed with TPL (Fig. 2a and b). A single whole-body exposure of mice to a lethal dose of ionizing radiation results in a plethora of syndromes.

At the cellular level, ionizing radiation can induce damage to biologically important macromolecules such

as DNA, proteins, lipids and carbohydrates in various organs. There has been no study in literature that describes prevention of radiation-induced DNA damage by TPL and this study is an advancement of knowledge in this regard. Our results show that TPL significantly prevented DNA damage induced by radiation using alkaline single cell gel electrophoresis, which is a sensitive technique to measure single strand breaks. It was also found that TPL feeding significantly reduced tail length, tail moment and % DNA in tail, which are parameters to indicate DNA damage in leukocytes and spleen cells of mice subjected to whole body γ -irradiation of 7.5 Gy. In summary, we have shown that TPL exerts its radio-protective abilities by modulating the activities of radiation-sensitive enzymes and protecting DNA from radiation induced damage. These results may prove useful in developing TPL based radioprotection regime. However, further work needs to be done to optimize the doses needed for application in clinical and other radiation exposure situations.

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